

Aman Singh

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Portfolio: amandev17.github.io/myportfolio | Location: Pune, India

Career Objective

Software Engineer with a backend and systems focus, experienced in designing scalable APIs, data-intensive pipelines, and database-driven applications using SQL, NoSQL, Redis, Django, and FastAPI. Comfortable working through full problem decomposition — algorithm design, performance tuning, and clean system architecture — not just gluing frameworks together.

Professional Experience

CodeAlpha — Java Intern

Jan 2026 – Feb 2026

- Architected high-concurrency Java applications simulating production-grade transactional workflows using OOP principles and JUnit testing.
- Engineered layered, decoupled system components to optimize codebase scaling and maintain strict separation of concerns.
- Optimized backend schema relations and computational data flows, mitigating runtime overhead and boosting request execution performance.

codeXslinger – Machine Learning Intern

Apr 2025 – Jun 2025

- Developed CNN-based computer vision pipelines and classification models, achieving a 90%+ predictive accuracy metric.
- Built automated machine learning pipelines for end-to-end preprocessing and data transformation, reducing manual data processing overhead by 60%.
- Managed project iteration life cycles using Agile scrum methodologies and tracked milestone sprints dynamically inside JIRA.

Projects

DocDedup – Multi-Format Document Deduplication Pipeline

2026

- Built a Django pipeline that detects duplicate pages across PDFs, Word docs, and scanned/camera-photo images using 4 complementary perceptual hashes (wHash, dHash, rotation-invariant ring-hash, centre-crop hash), with hash-bucket indexing cutting an $O(n^2)$ comparison problem down to a sparse candidate set.
- Designed a cost-ordered gate cascade (Hamming distance → windowed SSIM → AKAZE keypoint matching) that rejects over 95% of non-duplicate pairs via cheap bitwise checks before invoking expensive CV matching, with a centre-crop mode for rotated camera photos.
- Clustered confirmed duplicates with a Union-Find graph using per-node edge pruning, then generated a structured JSON report with thumbnails and rejection-stage explanations for every page.

pyflashdb – Open Source Unified Database Connector Library

May 2026

- Built an extensible Python database abstraction module mapping structural commands across MySQL, PostgreSQL, and MongoDB.
- Designed an object-oriented Adapter software pattern enabling code-agnostic runtime hot-swaps between relation and document storage structures.
- Implemented rigorous integration suites using PyTest, tracking sub-millisecond execution times and efficient pool-connection scaling; surpassed 1,800 PyPI downloads in the most recent 50-day window.

Education

B.E., Computer Engineering | Indira College of Engineering and Management (SPPU)

2023 – Present

- GPA: 9.2/10 | Core Modules: DSA, DBMS, Operating Systems, Computer Networks, Software Project Management

XII th | PDEA's Shri Mhalsakant Junior College | Percentage: 66.33%

2023

X th | CMS English Medium School | Percentage: 83.60%

2021

Skills

Languages: Java, Python, C++, SQL, Bash | **Databases & Caching:** MySQL, PostgreSQL, MongoDB, Redis

Core CS: Data Structures & Algorithms, Object-Oriented Design, System Design, Problem Solving

Backend & Architecture: Node.js, Django, Flask, FastAPI, RESTful APIs, Microservices, Connection Pooling

Cloud & DevOps: Docker, AWS (S3, EC2), CI/CD Baselines, Postman, PyTest, JUnit, Git/GitHub

ML & Image Processing: scikit-learn, TensorFlow/Keras, OpenCV, Pillow, NumPy, PyMuPDF, Perceptual Hashing, SSIM

Methodologies: Agile, Scrum, SDLC Automation, Threat Risk Modeling, Cross-Functional Deliverables

Spoken Languages: English, Hindi, Marathi

Certifications

Developing Back-End Apps with Node.js and Express

Issued by: IBM | 2025

Foundations of Project Management

Issued by: Google | 2024

Achievements & Leadership

- 4th Semester Department Top Performer** — Indira College of Engineering and Management
- Open Source Architecture Contributor** — pyflashdb Core Development Team
- 1st Runner-up**, GDG Ideathon 2025 (backend architecture innovation)
- Letters of Recommendation** from leadership at both CodeAlpha and codeXslinger